Surrogate models for gravitational waves using a hp-greedy domain partitioning

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Outline of the presentation

1. Introduction
2. Surrogate models and hp-greedy approach
3. Software design: Scikit-ReducedModel
4. hp-greedy applied to gravitational waves (GW)
5. Conclusions

Introduction

- Why surrogate models for GW?
  There is a high computational cost of numerical simulations.
- Obtaining a single gravitational waveform using numerical relativity (NR) to solve the full Einstein equations can take months using supercomputers.
- Parameter estimation can require millions of sequential evaluations of GWs.
- Problems have been addressed with surrogate models:
  - Predictions in real time.
  - No loss of accuracy.

- We work with a parameterized function space $\mathcal{F} := h_\lambda(t),\ \lambda \in \mathcal{D}$.
- RB: a representative and low-dimensional basis of $\mathcal{F}$.
- RB built with high-fidelity training set $\mathcal{T} := \{h_{\lambda_i}(t)\}_{i=1}^m$
    1. Apply a Greedy algorithm: at each step, the worst function of $\mathcal{T}$ represented by the basis is added to it.
    2. End criteria: reach dimension n_{max} or representation error ϵ.

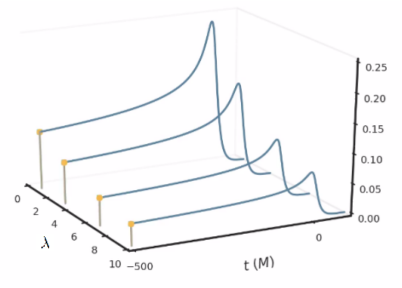


Figure 1: Schematic picture of Training set $\mathcal{T} := \{h_{\lambda_i}(t)\}_{i=1}^m$ [1].

Surrogate models: 1) Reduced basis with hp-greedy approach.

- Improve prediction time by partitioning the parameter space.
- Adaptive and recursive partitioning to the training set $\mathcal{T}$.
- Resulting tree structure allows fast predictions.

1. Build a global RB
2. Partition if RB does not reach tolerance ϵ within dimension n_{max}.
3. End of partitioning: Reach tolerance ϵ or tree depth l_{max}.

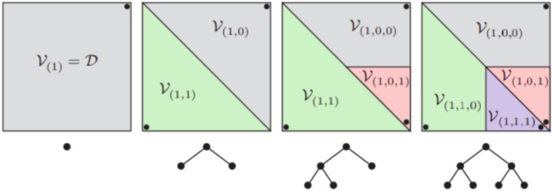


Figure 2: Schematic partitions and tree structure given by hp-greedy.

Surrogate models: 2) Empirical Interpolation Method (EIM)

- Using a reduced basis $\{e_i(t)\}_{i=1}^{n}$:
    - Build an expression for an *empirical interpolant* $\mathcal{I}[h_\lambda](t)$, $\forall \lambda \in \mathcal{D}$.
    - Select an optimal set of *empirical times* $\{T_i\}_{i=1}^{n}$ to interpolate.

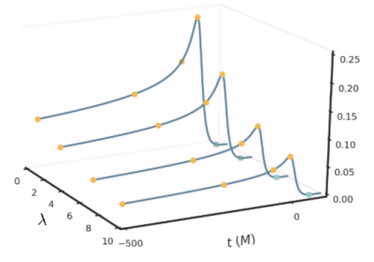


Figure 3: Yellow dots represent *empirical times* T_i.

Surrogate models: 3) Regressions through parameter space

- To evaluate an empirical interpolant $\mathcal{I}[h_\lambda](t)\ \forall\lambda$, we need $\{h_\lambda(T_i)\}_{i=1}^{n}\ \forall\lambda$.
- For each T_i we only know $h_{\lambda_j}(T_i)$, $\lambda_j \in \mathcal{T}$.
- For each T_i, train a Machine Learning (ML) algorithm to obtain $h_\lambda^{ML}(T_i), \forall\lambda$.

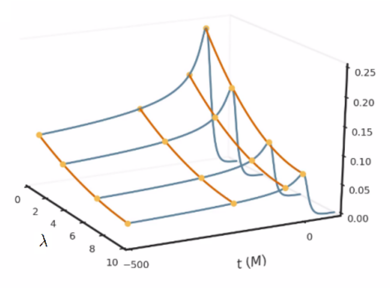


Figure 4: Orange curves: regressions for each *empirical time* T_i.

Surrogate models: Making a prediction

- Using trained regressions $\{h_\lambda^{\mathrm{ML}}(T_i)\}_{i=1}^n$, a prediction is given by

$$h_\lambda^{surr}(t) := \mathcal{I}[h_\lambda](t) = \sum_{i=1}^{n} B_i(t) h_\lambda^{\mathrm{ML}}(T_i).$$

- $B_i(t)$ are built in EIM training stage, before making predictions.

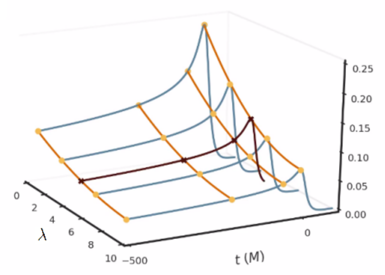


Figure 5: The black curve represents a prediction $h_\lambda^{surr}(t)$ for a given λ.

Software design

- We created the package Scikit-ReducedModel [2] that implements all stages of surrogate models and hp-greedy procedure.
    - Open source
    - Publicly available
- It was developed with a focus on *Scikit-Learn*:
    - Build custom models easily.
    - User-friendly, efficient, and well-documented interface.

Software design

Design principles:

- **Non-proliferation of classes.** Only the learning algorithms are represented with a Python class.
- **Composition.** If an algorithm is understood as a composition of other algorithms, express it as a composition of them.
- **Default values** for each algorithm.

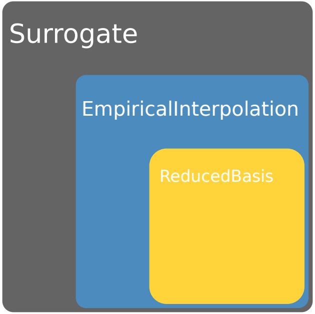
All objects share an API with three complementary interfaces:

- **Estimator.** Define the instantiation of objects with their respective hyperparameters and expose a `fit` method.
- **Transformer.** Receive input data and transform it, using the `transform` method.
- **Predictor.** Extend the notion of estimators, adding a `predict` method.

Software design

The public API of Scikit-ReducedModel consists of three classes: reducedbasis, empiricalinterpolation, and surrogate.

- A surrogate model is obtained after composing instances of each class.
- The modularization allows for specific studies with Reduced Basis or Empirical Interpolation Method.
- Installation: `pip install Scikit-ReducedModel`



hp-greedy applied to gravitational waves

- Compare dimensionality between a global Reduced Basis and hp-greedy bases [3, 4].
- Function space $h_\lambda(t)$: GWs emitted by two spinning but non-precessing black holes.
- $\lambda = (q, s_1, s_2), \quad q \in [1, 8], \; s_1 = s_2, \; s_i \in [-0.8, 0.8]$.

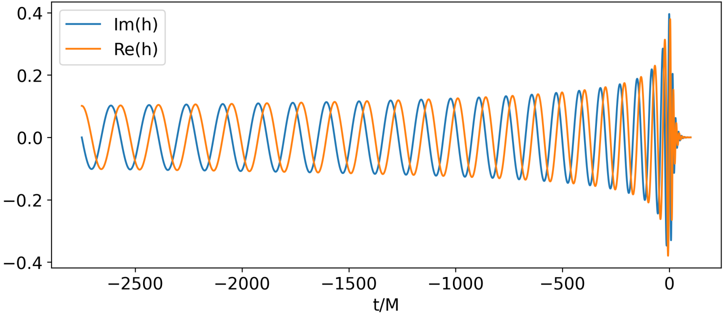


Figure 7: Sample of gravitational wave with $\lambda = (1, -0.8, -0.8)$

hp-greedy applied to gravitational waves

- hp-greedy training: Search of RB seed and l_{max} with Optuna, a hyperparameter optimization framework.
- Bases up to 4 times smaller with no loss of accuracy.

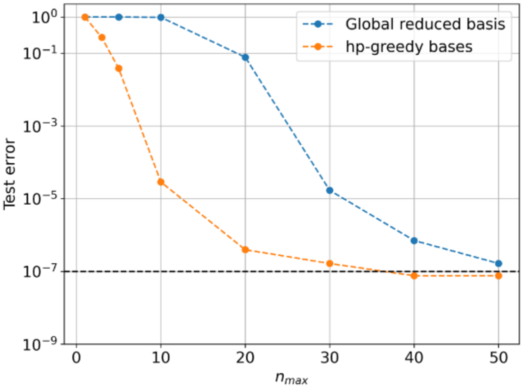


Figure 8: Test error vs n_{max}. $\lambda = (q, s_1, s_2), s_1 = s_2$

Conclusions

- `Scikit-ReducedModel` has been released. It is designed to build precise and fast approximations of gravitational waves.
- Designe based on the standards of `Scikit-Learn`, emphasizing easy usability.
- The package provides the possibility to conduct separate studies with each preprocessing step.
- We found high dependance of hp-greedy bases accuracy with l_{max} and the global RB seed.
- *hp-greedy* refinement allows faster GWs predictions with no loss of accuracy.

References

Varma, V. *Surrogate modeling.* `https://vijayvarma392.github.io/SurrogateMovie/`.

Cerino, F. y Rodriguez-Medrano, A. *scikit-reducedmodel.* `https://github.com/francocerino/scikit-reducedmodel`. 2023.

Cerino, F., Diaz-Pace, J. A. y Tiglio, M. An automated parameter domain decomposition approach for gravitational wave surrogates using hp-greedy refinement. *Classical and Quantum Gravity* **40,** 205003 (2023).

Cerino, F., Diaz-Pace, J. A., Tassone, E. A., Tiglio, M. y Villegas, A. Hyperparameter Optimization of an hp-Greedy Reduced Basis for Gravitational Wave Surrogates. *Universe* **10.** ISSN: 2218-1997 (2024).

Thanks!